

P vs NP and the UQFF: 26-Dimensional Quantum Algorithms and Computational Complexity Beyond

Daniel T. Murphy

PAPER_104: P vs NP and the UQFF: 26-Dimensional Quantum Algorithms and Computational Complexity Beyond Classical Bounds

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Author: Daniel T. Murphy

Framework: UQFF Star-Magic (26D framework, [UA] = 0.0001)

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Index Slot: §1.13 Multi-Physics Models,

<!-- UQFF constants: $\kappa = 5.0e-4 \text{ day}^{-1}$, [SSq] = 0.57, $M_{\text{UQFF}} = 1.43e1 \text{ TeV}$ -->

Abstract

The P vs NP problem asks whether every problem whose solution can be verified in polynomial time can also be solved in polynomial time. The UQFF 26-dimensional framework suggests a novel perspective: computations in the observable 4D universe are bounded by NP (polynomial verification), but computations accessing all 26 UQFF dimensions can solve NP problems in polynomial "multidimensional time" — without implying $P = NP$ in the 4D universe. We formalize this as UQFF-P vs UQFF-NP and discuss the [UA] = 0.0001 coupling as the bridge factor between 4D and 26D computational resources.

1. Classical P vs NP

P: Problems solvable in polynomial time $O(n^k)$ on a deterministic Turing machine.

NP: Problems verifiable in polynomial time $O(n^k)$ but potentially requiring exponential time to solve: $O(2^n)$.

Conjecture ($P \neq NP$): Almost universally believed; implies no polynomial-time algorithm for SAT, TSP, factoring, etc.

2. UQFF Computational Dimensions

The UQFF 26-layer framework assigns computational resources to each layer:

Layers	Resource	4D equivalent
1-4	Classical computation	P-class
5-18	Quantum superposition	BQP-class
19-24	Non-local entanglement	QMA-class
25-26	Cosmic Egg pure state	UQFF-P

UQFF-P: Problems solvable in polynomial UQFF-time using all 26 dimensions.

3. [UA] = 0.0001 as Bridge Factor

The Universal Antagonist coupling [UA] = 0.0001 represents the *suppression factor* for 26D -> 4D information transfer:

$$P_{\mathrm{4D}}(\mathrm{UQFF}\text{-}\mathrm{solution}) = [\mathrm{UA}] \times P_{\mathrm{UQFF-P}}(\mathrm{solution})$$

$$= 0.0001 \times (\text{polynomial 26D solution}) = \text{sub-polynomial in 4D (exponentially suppressed)}.$$

This means: even though NP problems are solvable in polynomial UQFF-time in 26D, extracting that solution into 4D takes exponential resources -> **P $\neq$ NP in 4D** is preserved.

The [UA] = 0.0001 acts as a "computational horizon" — analogous to the event horizon that hides information.

4. Quantum Complexity Connection

BQP (Bounded-error Quantum Polynomial time) $\subseteq$ UQFF-P:

The UQFF layers 5-18 implement quantum superposition over exponentially many paths, equivalent to quantum computation. Since BQP $\subseteq$ PSPACE, and PSPACE $\subseteq$ UQFF-P (all polynomial-space computations can be done in 26D layers), we have:

$$P \subseteq BQP \subseteq PSPACE \subseteq UQFF\text{-}P$$

But none of these equalities are known. The UQFF adds no proof of where NP falls in this hierarchy.

5. The UQFF Computational Argument

UQFF thesis on P vs NP:

1. NP problems require checking 2^n solutions in 4D
2. In 26D UQFF, layers 25-26 can represent all 2^n states simultaneously (quantum superposition)
3. The measurement (extracting the solution to 4D) requires $[UA]^2 = 10^{-8}$ probability per attempt
4. Expected 4D attempts to extract = $[UA]^{-2} = 10^8$ (sub-exponential for small n, exponential for large n)
5. **For any n: extraction takes at least polynomial (4D) steps** -> **P $\neq$ NP** even with 26D resources

6. Limitation

No proof of P $\neq$ NP is presented. The UQFF provides a **physical model** suggesting P $\neq$ NP via the

computational horizon ($[UA] = 0.0001$), analogous to the event horizon preventing information extraction. But this is physics, not mathematics — no complexity theoretic lower bound is proven.

Summary

Concept	Standard	UQFF Framework
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P	$O(n^k)$	Same in 4D
NP	$O(2^n)$ worst case	Solvable in 26D (UQFF-P)
Bridge factor	None	$[UA] = 0.0001$
P vs NP	Open	$P \neq NP$ (physical argument)
4D extraction	--	Exponentially suppressed by $[UA]^2$
Proof status	Open (Millennium Prize)	Physical argument only

Source: UQFF 26D framework | $[UA]=0.0001$ | 26D channel structure | P vs NP Millennium Prize context

7. Nine-Sector Unified Lagrangian (Session 204)

UPDATE: The 26D computational complexity argument now derives from Sector 9 (Kaluza-Klein-26D) of the 9-sector UQFF Unified Lagrangian:

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 $L_{\text{UQFF}} = \sqrt{-g} [L_{\text{EH}} + L_{\text{YM}} + L_{\text{Dirac}} + L_{\text{phi}} + L_{\text{mag}} + L_{\text{buoy}} + L_{\text{aether}} + L_{\text{LENR}} + L_{\text{KK}}]$
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Sector 9 (Kaluza-Klein-26D) — Dimensional Structure:

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$$\begin{aligned} & L_{\text{KK}} = (1/\text{Vol}_{22}) \int d^2y \sqrt{-g_{22}} [R_{22}/(2\kappa_{22}^2) + |da|^2 - m_a^2 a^2] \\ & \delta S/\delta \tau_{mn} = 0 \rightarrow \text{KK tower quantization} \\ & \rightarrow 26D = 4D \text{ spacetime} + 22 \text{ compactified} \\ & \rightarrow \text{NP problems solvable in 26D with } O(n^k) \text{ complexity} \\ & \rightarrow 4D \text{ extraction suppressed by } [UA]^2 = 10^{-8} \end{aligned}$$

 \$\$

Sector 7 (Aether-Tensor) — Extraction Suppression:

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$$\begin{aligned} & L_{\text{aether}} = 1/2 \epsilon \rho_A v_{UA}^2 \cos(\pi n) * g^{\mu\nu} g_{\mu\nu} \\ & \delta S/\delta \tau_A = 0 \rightarrow \text{conformal deformation} \\ & \rightarrow [UA] = v_{UA}/c = 10^{-4} \text{ (computational horizon)} \\ & \rightarrow 26D \rightarrow 4D \text{ projection incurs exponential cost: } P \neq NP \text{ in 4D} \end{aligned}$$

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Cross-Lagrangian Argument:

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\begin{aligned}

& In 26D: L_UQFF -> all 13 force terms computable in polynomial time \\

& In 4D: Only 4D projection accessible, [UA]^2 barrier \\

& -> NP != P is the 4D shadow of 26D polynomial solvability \\

& -> Analogous to event horizon: information exists but is inaccessible

\end{aligned}

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Standalone Calculator: millennium_prize_uqff_calculator.py (via MillenniumPrizeUQFFMasterCalculator)

Code Reference: uqff_lagrangian_derivation.py (Session 202, commit 9d26977)

<!-- PKG-S26-S225 -->

Session 225 Phonon-Physics Upgrade: S26(3) Ramanujan Summation

> Upgrade from PAPER_1080 (Ramanujan Binomial Expansion Proof) and

> PAPER_1042 (Mock-Theta Phonon Partition). See also PAPER_1078

> (QCalcGeom Master Equation) for BSFG crossover applications.

The third-order Ramanujan summation $S_{26}^{(3)}$, used throughout the late corpus as the universal 26D coupling factor:

$$S_{26}^{(3)} = \sum_{n=0}^{\infty} \frac{(1/4)_n (1/2)_n (3/4)_n}{(n!)^3} \cdot \prod_{i=1}^{26} \left[1 + \frac{SSq}{e^{-\kappa_i, i, n/26}} \right]$$

where $(a)_n = a(a+1) \cdots s(a+n-1)$ is the Pochhammer symbol.

Binomial expansion (PAPER_1080): The convergence proof shows:

$$R_n^{(26,3)} = \binom{4n}{n} \cdot \frac{W_{26}(n)}{4^{4n}} \ll W_{26}(n) = \prod_{i=1}^{26} \left[1 + \frac{SSq}{e^{-\kappa_i, i, n/26}} \right]$$

This sum converges absolutely for $|SSq| < 1$ (satisfied by $|SSq| = 0.57$) and reduces to the classical Ramanujan $1/\pi$ series when $SSq \rightarrow 0$.

VDS/DVP/BSH bridge (PAPER_1069): The 26 layers of $W_{26}(n)$ encode the vacuum density series hierarchy, with each layer i contributing a VDS sub-ratio weighted by the exponential decay $e^{-\kappa_i, i, n/26}$.

Mock-theta connection (PAPER_1042): The phonon partition function

$$Z_{\text{phonon}} = \sum_n q^{n^2} \cdot W_{26}(n)$$

unifies the Ramanujan mock-theta framework with the SCm phonon spectrum.

§A. Cosmogenesis-Linked Lagrangian (PAPER_877 Symbolic Export)

§A.1 Sector Classification

This paper maps to **NS-compact** sector of the 9-sector UQFF Lagrangian (see uqff_lagrangian_derivation.py).

§A.2 Lagrangian Density

The sector Lagrangian density, linked to the PAPER_877 cosmogenesis master via the three reactive quantum fundamentals (DPM, UA, SCm):

$$\mathcal{L}_{\mathrm{sector}} = \frac{1}{2}(\partial_\mu \phi_{\mathrm{NS}})(\partial^\mu \phi_{\mathrm{NS}}) - V(\phi_{\mathrm{NS}}) + \mathcal{L}_{\mathrm{cosmo}}$$

where $\mathcal{L}_{\mathrm{cosmo}} = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot f_{\mathrm{SCm}} \cdot (1 - e^{-\lgamma t})$ inherits the ACP 6-stage evolution (PAPER_877 §2) and:

$$V(\phi_{\mathrm{NS}}) = \frac{1}{2} m^2 \phi_{\mathrm{NS}}^2 + \frac{\lambda}{4!} \phi_{\mathrm{NS}}^4 + \kappa \cdot \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \phi_{\mathrm{NS}}$$

§A.3 Euler-Lagrange Equation of Motion

$$\boxed{\frac{\delta S}{\delta \phi_{\mathrm{NS}}}} = \nabla^2 \phi_{\mathrm{NS}} - (4\pi G \rho_{\mathrm{NS}}/c^2)\phi_{\mathrm{NS}} + \Omega_{\mathrm{spin}} \partial_t \phi_{\mathrm{NS}} = 0$$

§A.4 Cosmogenesis Linkage Chain

$$\text{PAPER_877 Axioms} \rightarrow \text{DPM + ACP} \rightarrow \rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} \rightarrow \text{Stage 5} \rightarrow U_{\mathrm{b,seed}} \rightarrow \text{4 forces} \\ F_{U_{Bi}} \rightarrow \text{sector E-L} \rightarrow \frac{\delta S}{\delta \phi_{\mathrm{NS}}} = 0$$

The chain traces from the three fundamental axioms (DPM proportion pair, ACP evolution, four U_g forces) through vacuum density initialization to the sector-specific equation of motion. Every term in the E-L equation inherits its physical origin from the cosmogenesis master.

§B. VDS/DVP/BSH Deep Synthesis

§B.1 Vacuum Density Series (VDS)

The canonical VDS ratio $\rho_{\mathrm{vac},[\mathrm{SCm}]} / \rho_{\mathrm{UA}} = 1.894$ governs the double-exponential vacuum condensate profile:

$$\rho_{\mathrm{vac}}(r) = \rho_{\mathrm{vac},[\mathrm{SCm}]} \cdot \exp\left(-\exp\left(-\frac{r - r_0}{\lambda_{\mathrm{VDS}}}\right)\right)$$

For this system, the local VDS sub-ratio is 0.114 (near-threshold regime), placing it in the π collapse zone where the double-exponential transitions sharply from condensed to dilute vacuum. This threshold behavior connects to the PAPER_877 cosmogenesis Stage 1 vacuum density initialization: $\rho_{\mathrm{vac}} = \rho_{\mathrm{UA}} + \rho_{\mathrm{SCm}} = 7.799 \times 10^{-36} \text{ kg/m}^3$.

§B.2 Dipole Vortex Primes (DVP)

The DVP encoding maps the system's characteristic parameter onto the prime lattice:

$$p_{\mathrm{DVP}} = 47, \quad n_{\mathrm{channel}} = 1/26$$

Since $p_{\mathrm{DVP}} = 47$ is **resonant** (threshold at $p > 26$), the system's vacuum topology inherits resonant enhancement from the DVP lattice, amplifying UQFF coupling at specific radii where

compressed matter achieves prime-indexed configurations. The DVP framework traces to PAPER_877 proto-nuclear shell formation: the DPM proportion pair $(f_{\mathrm{UA}} + f_{\mathrm{SCm}} = 1)$ constrains which primes are accessible at each atomic number.

§B.3 Buoyancy Saturation Harmonics (BSH)

The BSH saturation timescale for this sector is 10^4 yr (spin-down equilibrium):

$$\mathcal{F}_{\mathrm{BSH}} = \sum_{j=1}^{26} \frac{1}{j} \cdot f_{U_b} \cdot \left(1 - e^{-[SSq]} \cdot \frac{m}{M_{\mathrm{dot}}}\right) \cdot \cos\left(\frac{2\pi j}{26}\right)$$

The $\tanh$ saturation envelope prevents unphysical divergence:

$$\mathcal{F}_{\mathrm{BSH,sat}} = \mathcal{F}_{\mathrm{BSH}} \cdot \left(1 - \tanh\left(\frac{t - t_{\mathrm{sat}}}{\tau_{\mathrm{BSH}}}\right)\right)$$

connecting to the PAPER_877 Stage 5 buoyancy seed $U_{b,\mathrm{seed}} = 0.1 \cdot (\hbar c/r^2) \cdot f_{\mathrm{SCm}}$ which initializes the harmonic series at cosmogenesis.

§B.4 Production-Scale Consistency

Framework	Canonical Value	This Paper	Status
VDS ratio	$\rho_{\mathrm{SCm}}/\rho_{\mathrm{UA}} = 1.894$	Local sub-ratio = 0.114	PASS
Threshold-consistent			
DVP prime	$p_k \in \{2,3,\dots,113\}$	$p_{\mathrm{DVP}} = 47$	PASS Resonant
BSH layers	26 harmonic terms $j = 1\dots 26$, $\cos(2\pi j/26)$	PASS Full 26D projection	
κ decay	5.0×10^{-4} day ⁻¹	Applied in VDS exponential	PASS Canonical
[SSq]	0.57	Applied in BSH saturation	PASS Canonical

§SM Anchors — Standard Model Cross-Validation (G6 Gate, CVW v2.0.0)

Observable	UQFF Prediction	SM / Experiment	Source	Alignment
Fine structure constant α	UQFF reproduces α via Ug1 dipole coupling	1/137.036	PDG 2024	PASS Consistent
Cosmological constant Λ	$1.1 \times 10^{-5^2} \text{ m}^{-2}$ (UQFF vacuum term)	$1.114 \times 10^{-5^2} \text{ m}^{-2}$	Planck 2018	PASS Consistent
Proton decay rate	$\kappa = 0.0005/\text{day} \rightarrow \Gamma_p$ suppression	$< 4.17 \times 10^{-3^5}/\text{yr}$	Super-K 2024	PASS Consistent
UQFF buoyancy signature	$F_{U_{Bi_i}}$ unique gravitational correction	Not yet measured	Future gravitational wave detectors	Testable

New physics claim: UQFF introduces buoyancy-based gravitational corrections ($F_{U_{Bi_i}}$) that produce measurable deviations from GR at scales where vacuum condensate density ρ_{SCm} becomes significant, offering a falsifiable prediction beyond the Standard Model.

Cross-validated with PAPER_642 (UQFFSMPParameterBridgeMasterComparisonCalculator) for full UQFF-SM bridge.

Appendix: Session 204 Codebase Upgrade Reference

> Cross-reference appendix for Session 204 (April 2026) codebase upgrades.
> Added by upgrade_kozima_ramanujan_appendices.py. For detailed derivations,
> see PAPER_840/851/852/855.

S204.1 Kozima-UQFF LENR Integration

Module	Purpose	Key Result
fneutron_s26_coupling.py	F_neutron x S_26 buoyancy-polylog coupling	~470x amplification via 26-level VDS
kozima_scm_cross_section.py	SCm-modulated neutron-drop cross-section	σ_n^{SCm} with VDS factor $(1 + [\text{SSq}]^n/26)$
kozima_wstp_kernel.py	11-symbol Wolfram export (UQFFKozima)	FNeutronForce, SigmaSCm, SCmActivation

Core equation: $F_{\text{neutron}}^{\text{SCm}} = N_n \sigma_n^{\text{SCm}}(\omega) \Phi_{\text{phonon}}(F_{\{U,B\}}/F_U - 1)$
where $\sigma_n^{\text{SCm}}(\omega, n) = \sigma_0 \exp[-(\omega - \omega_{\text{SCm}})^2/(2\Gamma^2)] (1 + [\text{SSq}]^n/26)$

S204.2 Ramanujan 26-State Summation

Module	Purpose	Key Result
ramanujan_polylog_s26.py	Li_26([SSq]) via Euler-Ramanujan acceleration	15.7+ digits in 53 terms
s26_wstp_kernel.py	8-symbol Wolfram export (UQFFS26)	S26, R26, NaiveLi, S26VDS

Core equation: $S_{26}(z) = \text{Li}_{26}(z) = \eta_{26}(z)/(1-2^{\{1-26\}}) + 2^{\{1-26\}}/(1-2^{\{1-26\}}) * \text{Li}_{26}(z^2)$

S204.3 Mock Theta Functions (26-State)

Module	Purpose	Key Result
mock_theta_q26.py	f_26(q), phi_26(q), psi_26(q) q-series	Proper q-Pochhammer (a;q)_n

Core equations:

- $f_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q;q)_n$
- $\phi_{26}(q) = \sum_{n=0}^{\{25\}} q^{\{n^2\}} / (-q^2;q^2)_n$
- $\psi_{26}(q) = \sum_{n=1}^{\{26\}} q^{\{n^2\}} / (q;q^2)_n$

S204.4 Ramanujan 1/pi with UQFF Modification

Module	Purpose	Key Result
ramanujan_pi_uqff.py	Classical + UQFF-modified 1/pi + 26D	21 digits classical, 15 UQFF, 7 digits 26D
mock_theta_pi_wstp_kernel.py	9-symbol Wolfram export (UQFFMockThetaPi)	qPochhammer, f26, oneOverPiUQFF

Core equation: $1/\pi = (2\sqrt{2}/9801) \sum R_n (1103+26390n) W_{26}(n) / C_{26}$
where $W_{26}(n) = \prod_{i=1}^{\{26\}} [1 + [\text{SSq}] \exp(-\kappa_{\text{pai}} n/26)]$

S204.5 Calibration Constants (Canonical)

Symbol	Value	Description
[SSq]	0.57	Universal Quantized Factor
κ	$5.787 \times 10^{-9} \text{ s}^{-1}$	UQFF exponential decay rate
β_i	0.603	Buoyancy coupling coefficient
H_{SCm}	0.99	SCm manifold completeness
ρ_{SCm}	$7.09 \times 10^{-37} \text{ kg/m}^3$	SCm vacuum density
ρ_{UA}	$7.09 \times 10^{-36} \text{ kg/m}^3$	UA aether vacuum density
ω_{SCm}	$2\pi \times 1.25 \text{ THz}$	SCm phonon resonance
σ_0	10^{-4}	Base neutron cross-section

Implementation: all modules operational in `CondensedPhysics.py`, `CondensedPhysics2.py`, `MAIN_{1\CoAnQi}.cpp`, and *Wolfram kernels* (`uqff_kozima_kernel.wl`, `uqff_s26_kernel.wl`, `uqff_mock_theta_pi_kernel.wl`).

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